

AI3613 Database System Concepts Homework 4

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Problem 1

1. A

$500,000/200 = 2500$ runs in the first pass, and $\lceil 2500/199 \rceil = 13$ runs in the second pass, and $\lceil 13/199 \rceil = 1$ is the final sorted file that we do not count. So the total number of runs is $2500 + 13 = 2513$.

2. C

Pass 0 for pre sort, pass 1 to merge and pass 2 to get the final sorted file. So the total number of passes is 3.

3. C

After pass 0, there are 2500 sorted blocks with 200 pages each. After pass 1, there are 13 sorted blocks, 12 of them should have $199 \cdot 200$ pages and the rest one have rest of the pages.

4. B

In each pass, we read all and write all. So the total number of I/Os is $500,000 \cdot 6 = 3,000,000$.

Problem 2

1. a. D

For partion, we need read in and out for all.

b. B

Need to read all partions once but output is ignored.

2. C

$P(R) + P(R) * P(S) = 1081800$, $P(R) = 1800$, $P(S) = 600$. No way. We are to use 35 buffer pages to store outer table. $P(R) + P(R) * \lceil P(S)/(B - 1) \rceil = 33000$

3. C

Switch R and S. The result is the same by coincidence.

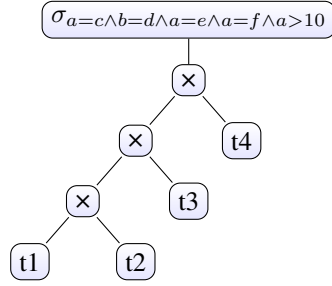
4. a. C

Only a sequential scan.

b. A

Suppose $a=0$, we need to join every tuple: the same cost as nested loop join.

Problem 3

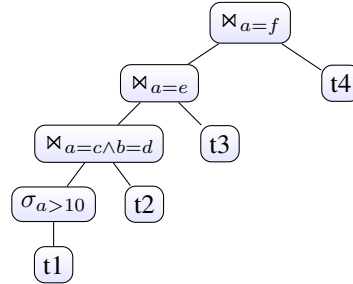


$$\sigma_{a=c \wedge b=d \wedge a=e \wedge a=f \wedge a > 10} (t1 \times t2 \times t3 \times t4),$$

$$\sigma_{a > 10}(\sigma_{a=c \wedge b=d}(\sigma_{a=e}(\sigma_{a=f}(t1 \times t2 \times t3 \times t4))))$$

Use theta join to replace cross join:

$$(\sigma_{a > 10}(t1) \bowtie_{a=c \wedge b=d} t2) \bowtie_{a=e} t3 \bowtie_{a=f} t4$$

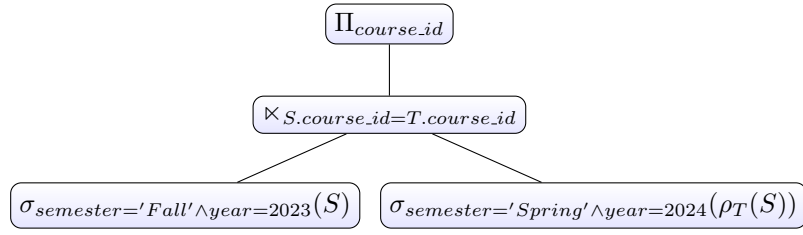


Problem 4

1. The query can be formulated as

$$\Pi_{course_id} (\sigma_{semester='Fall' \wedge year=2023}(S) \bowtie_{S.course_id=T.course_id} \sigma_{semester='Spring' \wedge year=2024}(\rho_T(S)))$$

The query plan is as follows:



2. Similar to the given query, antijoin indicates not exists. The query

$$\Pi_{\text{gpa}}(\text{Students} \bar{\bowtie}_{\text{Students.stu.id=Punished.punished_stu.id}}(\text{Punished}))$$

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1 SELECT S.gpa
2 FROM Students as S
3 WHERE S.stu_id NOT IN (
4     SELECT P.punished_stu_id
5     FROM Punishment as P
6 );

```

Algorithm 1: $R \bar{\bowtie}_{\theta} S$

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1 for each page  $P_r$  in  $R$  do
2     for each page  $P_s$  in  $S$  do
3         for each tuple  $t_r$  in  $P_r$  do
4             for each tuple  $t_s$  in  $P_s$  do
5                 if  $\theta(t_r, t_s)$  then
6                     break
7                 add  $t_r \bar{\bowtie} t_s$  to the result

```

Problem 5

How long does it take you to finish the assignment? Give a score (1,2,3,4,5) to the difficulty of each problem. List all your collaborators if you have any.

Abut 3 hours including review. 2, 2, 3, 3.